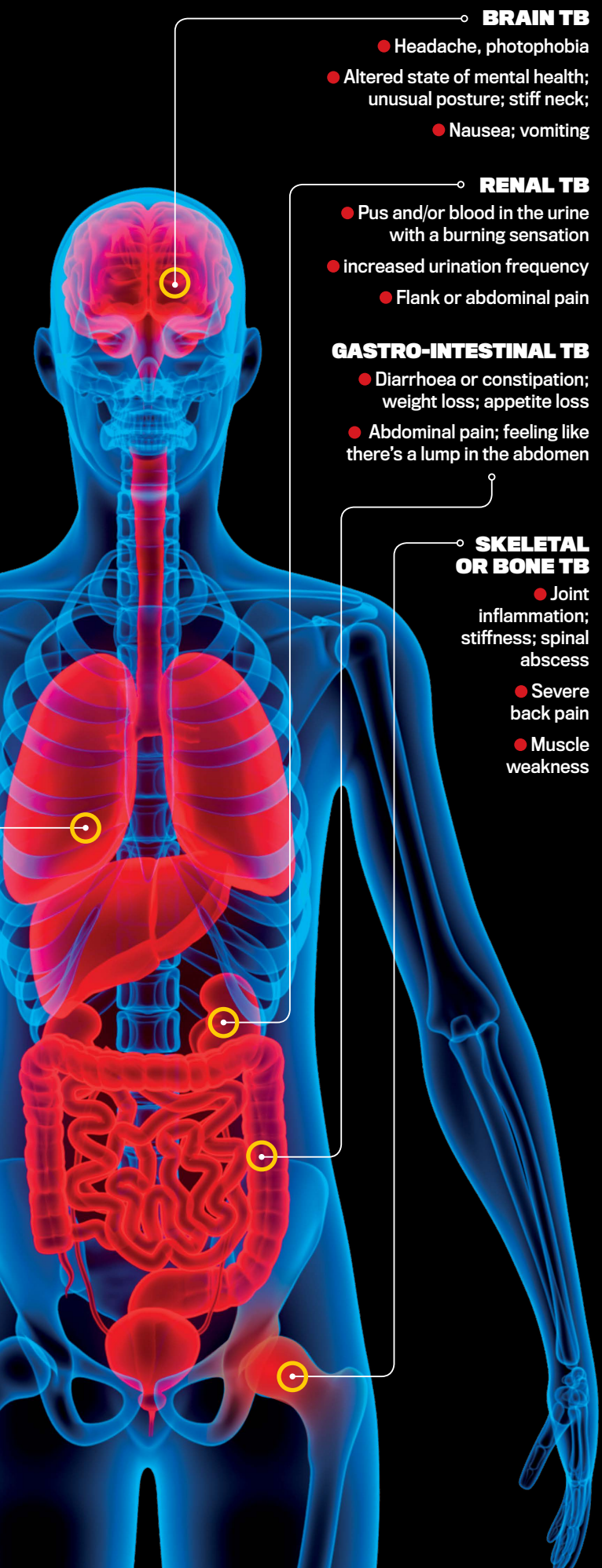


BRAIN TB

- Headache, photophobia
- Altered state of mental health; unusual posture; stiff neck;
- Nausea; vomiting

RENAL TB

- Pus and/or blood in the urine with a burning sensation
- increased urination frequency
- Flank or abdominal pain

GASTRO-INTESTINAL TB

- Diarrhoea or constipation; weight loss; appetite loss
- Abdominal pain; feeling like there's a lump in the abdomen

SKELETAL OR BONE TB

- Joint inflammation; stiffness; spinal abscess
- Severe back pain
- Muscle weakness

TB THERAPIES IN INDIA

STANDARD TB

Drugs: Isoniazid, rifampicin, pyrazinamide and ethambutol

Duration ~ 24 weeks (initial phase of eight weeks consists of the four first-line drugs, followed by a continuation phase of another 16 weeks of isoniazid, rifampicin and ethambutol

Common side-effects: Chest pain, fatigue, fast heartbeat, coughing up blood, nausea

Cost: Rs 7,500 in public sector; Rs 20,000 in private sector

EXTENSIVELY DRUG-RESISTANT (XDR) TB

Drugs: XDR TB is resistant to one or more drugs used in MDR-TB and treatment fails with the wrong drug, dosage or duration. A later generation of fluoroquinolones, with category IV and V injectible drugs such as clofazimine, linezolid, amoxicillin/ clavulanate, thioacetazone, imipenem/ cilastatin, high-dose isoniazid and clarithromycin are used

Duration: Can vary from case to case

Side-effects: Joint pain, skin discoloration, diarrhoea, depression, accumulation of blood in brain, insomnia, severe weakness

Cost: The above-mentioned study says the average treatment cost for pulmonary XDR TB is $8,401 (Rs 6.9 lakh)

MULTI-DRUG-RESISTANT (MDR) TB

Drugs: Kanamycin, levofloxacin, cycloserine, ethionamide, pyrazinamide and ethambutol. P-aminosalicylic acid is given as a reserve drug in case of intolerance of, or a reaction to, any one of the other drugs

Duration ~ 24-27 months (six-nine months of intensive phase, followed by a continuation phase of 18 months)

Common side-effects: Hearing loss, depression, psychosis, GI disturbance, blurred or changed vision, darkening of skin, fatigue and body ache

Cost: Per a 2019 study of 50 patients at Hinduja Hospital in Mumbai, average treatment cost for pulmonary MDR TB was $5,723 (Rs 4.7 lakh)

NEW ORAL TABLETS

↘ **Bedaquiline:** It kills the bacteria and stops its growth by inhibiting the production of an enzyme essential to its survival

Duration: Six months

Side-effects: Chest pain, dizziness, blood in the sputum, irregular heartbeat

↘ **Delamanid:** It blocks the production of two essential components of the bacteria's cell walls, causing its death.

Duration: Six months

Side-effects: Relatively fewer, but it does cause headaches, dizziness, nausea

**While bedaquiline, patented by Janssen Pharmaceuticals until July this year, was launched in the Indian market at Rs 7 lakh per vial, delamanid, patented by Otsuka Pharmaceuticals until October, costs about Rs 1.5 lakh per vial. Both are available only through government channels currently and are provided free of cost. The MSF estimates that in comparison, generic prices may be just around $8-$16 (Rs 658-Rs 1,316) per month*